MUSTAFA AVCU

SOFTWARE ENGINEER | AI/ML | DATA PLATFORMS | AWS

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PROFESSIONAL SUMMARY

Software engineer with 4 years of experience at Amazon building scalable backend services and event-driven data processing pipelines across advertising and customer service chatbot domains. Hands-on AWS experience with strong Python, Java, Scala, and TypeScript skills. Proven track record designing core backend architectures and optimizing high-scale systems, alongside specialized experience architecting production-grade MLOps infrastructure and safety guardrails for LLM-powered chatbot.

TECHNICAL SKILLS

Languages: Python, Java, Scala, TypeScript, SQL, JavaScript, C++
Cloud & Infrastructure: AWS, CDK, CloudFormation, Lambda, Step Functions, S3, DynamoDB, SQS, SNS
Data Engineering: Glue, Spark, EMR, ETL data pipelines, AWS Clean Rooms
AI/ML: Large Language Models (LLMs), Generative AI, guardrails, Keras, TensorFlow, SageMaker, Bedrock
Engineering: CI/CD, Docker, Git, infrastructure as code, microservices, REST APIs, on-call support

PROFESSIONAL EXPERIENCE

Amazon | Software Development Engineer II

Seattle, WA & Toronto, ON | August 2022 – June 2026

- Implemented a guardrail layer in a new LLM orchestration service connecting generative AI models to Amazon's customer service chatbot; validated customer inputs and model outputs to ensure safe, appropriate interactions, enabling the launch of Amazon's first LLM-powered customer support experience.
- Designed and built reusable machine learning infrastructure using Amazon SageMaker and automated CI/CD pipelines to streamline end-to-end model training, testing, and deployment; deployed customer issue-prediction models and reduced average chatbot interaction time by 9% (440 seconds to 400 seconds) with no negative impact on other metrics.
- Owned Retailer Ad Service's attribution integration with Amazon Demand-Side Platform (ADSP) to support display and video ads alongside Sponsored Products; migrated AWS Clean Rooms analytics engines from Redshift to Spark SQL ahead of deprecation deadlines to lift data volume caps, and built unified ETL data pipelines and attribution logic to jointly perform attribution across both ad programs.
- Redesigned the Retailer Ad Service's catalog ingestion architecture to support catalogs with 2M+ offers refreshed every two hours; eliminated throttling by optimizing Lambda concurrency to improve ingestion reliability and resolved high upsert latencies (300ms+) in catalog service by optimizing change-detection workflow logic.
- Authored comprehensive Operational Readiness Reviews (ORR) and system architecture diagrams for catalog scaling updates, driving 11 technical remediation tasks; reduced OpenSearch ingestion traffic by 90% via DynamoDB change-event filtering, winning a production impact award at an organization-wide hackathon.

Agency for Science, Technology and Research (A*STAR)

Singapore | Data Science Intern | June 2018 – September 2018

- Developed SeizNet, a convolutional neural network for seizure detection using EEG data from 29 patients; published at IEEE ICASSP 2019 and presented in Brighton, UK.
- Achieved 95.8% sensitivity at 0.17 false alarms per hour; a 2-channel configuration achieved 93.3% sensitivity and outperformed a traditional SVM baseline using all EEG channels.

EDUCATION

University of Southern California — M.S. in Computer Science | CGPA: 3.96/4.00 | May 2022
Selected coursework: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing

Bilkent University — B.S. in Electrical and Electronics Engineering | CGPA: 3.93/4.00 | June 2019

HONORS & AWARDS

- Two-time Amazon organization-wide hackathon winner among 20+ competing teams
- Annenberg Fellowship with admission to the USC PhD program (2019)
- IEEExtreme 10.0 (2016): 2nd in Turkey, 20th in IEEE Region 8, 65th global
- IEEExtreme 9.0 (2015): 3rd in Turkey, 30th in IEEE Region 8, 82nd global
- Nationwide University Entrance Exam (LYS): ranked in the top 300 out of 2 million students in Turkey (2014)

PUBLICATION

M. T. Avcu, Z. Zhang, and DWS. Chan, "Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks" IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.